

□ Hardware Requirements - Complete Technical Specification

This document provides comprehensive hardware specifications, platform details, and technical references for the TM4C1294-PWM-Controller system.

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TM4C1294XL Platform Overview

EK-TM4C1294XL Connected LaunchPad

The Texas Instruments EK-TM4C1294XL is a comprehensive development platform featuring the powerful TM4C1294NCPDT microcontroller. This platform provides the ideal foundation for high-performance embedded applications requiring precise timing and comprehensive connectivity.

Official Product Information

- **TI Product Page:** [EK-TM4C1294XL LaunchPad](#)
- **User Guide:** [SPMU373C - EK-TM4C1294XL User's Guide](#)
- **Schematic:** [EK-TM4C1294XL Design Files \(Rev D\)](#)

Core Microcontroller: TM4C1294NCPDT

Processor Specifications

- **CPU Core:** ARM Cortex-M4F with floating-point unit
- **Operating Frequency:** 120 MHz maximum
- **Architecture:** 32-bit RISC with Thumb-2 instruction set
- **Performance:** 150 DMIPS at 120 MHz
- **Floating Point:** IEEE 754 single-precision FPU

Memory Configuration

- **Flash Memory:** 1024 KB (1 MB) program memory
- **SRAM:** 256 KB total system RAM
- **EEPROM:** 6 KB non-volatile data storage
- **ROM:** 128 KB with TivaWare DriverLib and boot loader

Memory Map Detail

Flash Memory: 0x00000000 - 0x000FFFFF (1024 KB)
SRAM: 0x20000000 - 0x2003FFFF (256 KB)
Bit-band SRAM: 0x22000000 - 0x23FFFFFF (32 MB alias)
Peripherals: 0x40000000 - 0x5FFFFFFF
Private Periph: 0xE0000000 - 0xE00FFFFF
EEPROM: 0x400AF000 - 0x400AFFFF (6 KB)

Official TI Documentation References

Primary Datasheets

- [TM4C1294NCPDT Datasheet](#): Complete microcontroller specifications
- [TivaWare Peripheral Driver Library User's Guide](#): Software library documentation
- [ARM Cortex-M4 Technical Reference Manual](#): Core processor details

Application Notes

- [System Design Guidelines for TM4C129x Family](#): Hardware design best practices
 - [TM4C129x Ethernet Applications for lwIP](#): Network connectivity implementation
 - [TM4C129x Ethernet Applications for TI-RTOS NDK](#): Real-time network applications
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Detailed Hardware Specifications

Power Requirements

Supply Specifications

- **Operating Voltage**: 3.3V nominal (3.0V to 3.6V range)
- **I/O Voltage**: 3.3V (some devices/pins may be 5V tolerant on *inputs*; treat pins as 3.3V unless verified)
- **Power Consumption**:
 - Active mode (120 MHz): ~140 mA typical
 - Sleep mode: < 1 mA
 - Deep sleep: < 500 μ A

LaunchPad Power Options

1. **USB Power**: Via micro-USB connector (5V $\rightarrow$ 3.3V regulation)
2. **External Power**: 7-15V DC via barrel jack (regulated to 3.3V)
3. **BoosterPack Power**: 3.3V directly to BoosterPack connector

Clock System

Primary Clock Sources

- **Main Oscillator**: 25 MHz crystal (Y1 on LaunchPad)
- **Internal Oscillator**: 16 MHz $\pm$ 1% precision

- **RTC Oscillator:** 32.768 kHz for real-time clock (optional)

PLL Configuration

- **Input Range:** 5-25 MHz
- **VCO Range:** 320-480 MHz
- **Output:** Divided to achieve 120 MHz system clock
- **Jitter:** < 150 ps RMS

```
// Clock configuration for 120 MHz operation
SysCtlClockFreqSet((SYSCTL_XTAL_25MHZ | SYSCTL_OSC_MAIN |
                    SYSCTL_USE_PLL | SYSCTL_CFG_VCO_480), 120000000);
```

GPIO Capabilities

Port Configuration

- **Total GPIO Pins:** 90 pins across Ports A-Q
- **Drive Strength:** 2mA, 4mA, 8mA options
- **Special Features:**
 - 5V tolerant inputs (selected pins; verify per datasheet)
 - Open-drain outputs
 - Internal pull-up/pull-down resistors
 - Interrupt capability on all pins

Port Distribution on LaunchPad

- **Port F:** User LEDs (PF0-PF4) and switches (PF0, PF4)
- **Port N:** RGB LED (PN0-PN1)
- **Port J:** User switches (PJ0-PJ1)
- **Port Q:** Additional I/O and special functions

Pin Assignments and Connections

Project pinning (what this firmware actually uses)

This section documents the *actual* signal wiring used by the current firmware.

Signal	MCU pin	Peripheral / function	Notes
PWM output (to PSU)	PF2	M0PWM2 (PWM0 module, output 2)	Target is ~24.9kHz (TARGET_PWM_FREQ_HZ 24900U), duty via PSYN / PHASE*. 3.3V logic.

Signal	MCU pin	Peripheral / function	Notes
PWM input sense (from PSU)	PF3	Intended T1CCP1 (Timer1B capture), fallback GPIOF both-edge ISR	PWMIN reports f + duty at 1Hz on UART0. PF3 is not a user LED pin on this board (LEDs are PF0/PF4, PN0/PN1).
TACH input (fan tach sense)	PF1	GPIO interrupt (falling edge + reject filter)	Uses internal weak pull-up; treat as 3.3V only unless conditioned.
TACH output (fake tach to PSU)	PM3	Timer-based tach synthesizer	Driven by PHASE*, TSYN BOOT, TSYN COPY.
UART0 (ICDI debug/log)	PA0/PA1	U0RX/U0TX	Shows as /dev/ttyACM* on Linux.
UART3 (user command port)	PA4/PA5	U3TX/U3RX	Typically via external USB-serial adapter (/dev/ttyUSB*), 3.3V logic.
Session detect (DTR)	PQ1	GPIO input	Used to detect terminal connect/disconnect boundary.

UART Interface Pins

UART0 (ICDI Debug Interface)

- **TX:** PA1 (connected to ICDI via U5 level shifter)
- **RX:** PA0 (connected to ICDI via U5 level shifter)
- **Connector:** USB micro-B (virtual COM port)
- **Isolation:** Galvanically isolated via ICDI circuitry

UART3 (External Command Interface)

- **TX:** PA4 (BoosterPack pin 12 - J2)
- **RX:** PA5 (BoosterPack pin 11 - J2)
- **Voltage:** 3.3V CMOS logic levels
- **Protection:** Series resistors R9/R10 (33Ω each)

```
// UART3 pin configuration
GPIOPinConfigure(GPIO_PA4_U3TX);
GPIOPinConfigure(GPIO_PA5_U3RX);
GPIOPinTypeUART(GPIO_PORTA_BASE, GPIO_PIN_4 | GPIO_PIN_5);
```

Host-side device nodes: /dev/ttyUSB0 vs /dev/ttyUSB1 Linux assigns /dev/ttyUSB* numbers by enumeration order, so the UART3 adapter can appear as /dev/ttyUSB0 on one host and /dev/ttyUSB1 on another.

Recommended solutions:

- 1) Use stable names under /dev/serial/by-id/ (best).
 - `ls -l /dev/serial/by-id/`
 - Then pass that path into the tooling (examples):
 - `make capture UART3_DEV=/dev/serial/by-id/<your-uart3-adapter>`
 - `python3 tools/uart_session.py --uart3 /dev/serial/by-id/<your-uart3-adapter>`
- 2) Override Makefile variables per invocation (simple):
 - `make capture UART3_DEV=/dev/ttyUSB0`
 - `make auto UART3_DEV=/dev/ttyUSB1`

This avoids editing the Makefile per host.

Session Detection Pin

DTR Detection: PQ1

- **Function:** Data Terminal Ready input
- **Location:** BoosterPack expansion area
- **Configuration:** Input with internal pull-up
- **Logic:** Active-low DTR signal detection

Debug and Programming Interface

JTAG/SWD Connections (via ICDI)

- **SWDIO:** PC1 (Serial Wire Debug I/O)
 - **SWCLK:** PC0 (Serial Wire Debug Clock)
 - **TDO:** PC2 (Test Data Out)
 - **TDI:** PC3 (Test Data In)
 - **nTRST:** PC7 (Test Reset)
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Peripheral Utilization

Timer Modules

PWM0 module - PWM generation (PF2) In the current firmware, PWM output is produced by the **PWM peripheral** (not a TimerA PWM mode).

- **Pin:** PF2 (M0PWM2)
- **Module / generator:** PWM0, PWM_GEN_1, output PWM_OUT_2
- **Base frequency target:** 24.9 kHz (TARGET_PWM_FREQ_HZ 249000)
- **Duty cycle range:** 5–96% (enforced by command layer)
- **Clocking:** PWM_SYSCLK_DIV_1 (uses system clock)

Representative configuration (from main.c):

```
GPIOPinConfigure(GPIO_PF2_M0PWM2);
GPIOPinTypePWM(GPIO_PORTF_BASE, GPIO_PIN_2);
```

```
PWMClockSet(PWM0_BASE, PWM_SYSCLK_DIV_1);
```

```
PWMGenConfigure(PWM0_BASE, PWM_GEN_1, PWM_GEN_MODE_DOWN | PWM_GEN_MODE_NO_SYNC);  
PWMGenPeriodSet(PWM0_BASE, PWM_GEN_1, period);  
PWMPulseWidthSet(PWM0_BASE, PWM_OUT_2, pulse);  
PWMOutputState(PWM0_BASE, PWM_OUT_2_BIT, true);  
PWMGenEnable(PWM0_BASE, PWM_GEN_1);
```

Note: with a 120 MHz system clock, the period is quantized to an integer number of clock ticks. The firmware targets 24.9 kHz closely by computing `period = round(sysclk / 24900)`.

Available Additional Timers

- **Timer1-7:** 32-bit general purpose timers
- **Wide Timer0-5:** 64-bit timers for extended timing
- **Watchdog Timer:** System reliability monitoring

UART Modules

UART0 - Debug Output

- **Base Address:** 0x4000C000
- **Interrupt:** UART0_IRQn (21)
- **DMA:** DMA channel assignment available
- **FIFO:** 16-byte transmit/receive FIFOs

UART3 - Command Interface

- **Base Address:** 0x4000F000
- **Interrupt:** UART3_IRQn (24)
- **Features:** Full modem control signals available
- **Baud Rate:** Up to 1.8 Mbps theoretical maximum

ADC Capabilities (Available for Future Enhancement)

ADC0 Module

- **Resolution:** 12-bit (4096 steps)
- **Sampling Rate:** Up to 2 MSPS
- **Input Channels:** 20 external + 4 internal
- **Reference:** Internal 3.0V or external VREFA

ADC1 Module

- **Identical Specifications:** Dual ADC for simultaneous sampling
- **Digital Comparators:** 8 programmable comparators
- **Temperature Sensor:** Internal temperature monitoring

External Hardware Requirements

Power Supply Considerations

Recommended External Power

- **Voltage:** 9-12V DC for optimal regulation efficiency
- **Current:** Minimum 500mA for full-speed operation
- **Connector:** 2.1mm center-positive barrel jack
- **Regulation:** On-board LM3671 (3.3V/600mA switching regulator)

USB Power Limitations

- **Current:** Limited to 500mA USB specification
- **Performance:** May affect high-speed operation under heavy load
- **Recommendation:** External power for production applications

External UART Interface

Level Shifting Requirements For interfacing with RS-232 devices: - **IC Recommendation:** MAX3232 or equivalent - **Supply:** 3.3V operation - **Capacitors:** 4x 100nF ceramic for charge pump

USB-to-Serial Converter For development convenience: - **Recommended:** FTDI FT232R or CP2102-based modules - **Voltage:** 3.3V logic level compatibility - **Connections:** - TXD → PA5 (UART3 RX) - RXD → PA4 (UART3 TX) - GND → GND

PWM Output Interface

Direct 3.3V Logic Interface

- **Drive Current:** 8mA maximum per pin
- **Logic Levels:**
 - VOH: 2.4V minimum @ IOH = -2mA
 - VOL: 0.4V maximum @ IOL = 2mA

Power Driver Interface For high-current loads: - **Buffer IC:** 74HC244 or similar CMOS buffer - **MOSFET Driver:** For switching applications - **Isolation:** Optocouplers for electrical isolation

Development Tools

Programming and Debug Tools

Integrated Circuit Debug Interface (ICDI)

- **Functionality:** On-board JTAG/SWD debugger
- **Processor:** TM4C123GH6PM dedicated debug MCU
- **Interface:** USB 2.0 Full Speed
- **Compatibility:** OpenOCD, Code Composer Studio, Keil µVision

External Debug Probes (Optional)

- **SEGGER J-Link:** Professional debugging solution
- **TI XDS110:** High-performance TI-specific probe
- **OpenOCD Compatible:** Various ARM-based debug probes

Software Development Environment

Recommended Toolchains

1. **ARM GCC Toolchain**
 - Version: arm-none-eabi-gcc 10.3.1 or newer
 - Optimization: -Os for size, -O2 for performance
 - Debugging: -ggdb for GDB compatibility
2. **Code Composer Studio (CCS)**
 - TI's official IDE
 - Integrated debugging and profiling
 - Free for evaluation and educational use
3. **Keil μ Vision MDK-ARM**
 - Professional ARM development environment
 - Advanced debugging and trace capabilities
 - Commercial licensing required

Build System

- **Make:** Standard GNU Make with custom Makefile
 - **CMake:** Modern build system (future consideration)
 - **TivaWare:** Required peripheral driver library
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Performance and Timing Specifications

Real-time Performance

- **Interrupt Latency:** < 16 clock cycles (133ns @ 120MHz)
- **Context Switch:** < 100 clock cycles typical
- **PWM Update:** < 1ms from command to output change

Environmental Specifications

- **Operating Temperature:** -40°C to +85°C (industrial grade)
- **Storage Temperature:** -55°C to +150°C
- **Humidity:** 5% to 95% non-condensing
- **Vibration:** Per MIL-STD-883 (LaunchPad level)

Electrical Characteristics

- **ESD Protection:** ± 2 kV HBM, ± 200 V MM

- **Latch-up Immunity:** $\pm 200\text{mA}$
- **Input Leakage:** $\pm 1\mu\text{A}$ maximum @ 3.6V

This comprehensive hardware specification provides the complete technical foundation for understanding and extending the TM4C1294-PWM-Controller system. The EK-TM4C1294XL LaunchPad serves as an excellent development platform with extensive peripheral capabilities and professional development tool support.